

# Khansa Khanam

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## Research Interests

AI bias and fairness, benchmarking AI systems for minority and multilingual contexts, AI adaptability, and AI and the future of work. My work engineers around a consistent finding across projects: models are built for the median user, and everyone else absorbs the error — I focus on making models adaptable to the people using them (multilingual, minority, low-resource) and cheap enough to run that the organizations serving those people can actually own and govern them. Eager to bring this mixed-methods, applied research background to a PhD collaboration in these areas.

## Education

### M.S., Computer Science

University at Buffalo (SUNY)

Jan 2024 – Jun 2025

Buffalo, NY

### B.Tech., Information Technology

Anna University

Aug 2017 – Jul 2021

Chennai, India

## Research Experience

### Research Assistant

Rutgers University, School of Communication and Information

Jan 2026 – Present

New Brunswick, NJ

Advised by Tawfiq Ammari and Robert Wolfe

#### Documentation bias in clinician notes:

- Investigating whether patient race is recoverable from the writing style of clinician notes, as a proxy measure for documentation bias, across fine-tuned, instruction-tuned, and classical approaches.
- Fine-tuned open-weight models (Llama-2-7B, Bode-7B-PT) under imbalanced and class-balanced regimes; diagnosed the failure mode as architectural rather than tuning-related, with over half of records exceeding the 4K context window.
- Built an in-context-learning pipeline (Llama-3.1-8B with recursive semantic chunking and metadata-/TF-IDF-similarity-based few-shot retrieval), with strict pre-split partitioning to prevent leakage and majority-vote aggregation across chunks.
- Ran an ablation across approaches; a hybrid TF-IDF + BERTopic + SGDClassifier ensemble reached 69.8% accuracy / 0.67 macro-F1, outperforming every LLM configuration tested. Applied SHAP with a log-odds link to trace which semantic topics drove each class prediction.

#### Multilingual SDOH extraction from EHRs:

- Designed and ran an evaluation of Llama-3.3-70B Instruct for extracting Social Determinants of Health from Brazilian Portuguese clinical notes, running inference on untranslated source text to preserve culturally specific phrasing and to keep inference local rather than exposing patient data to cloud APIs.
- Built the prompt architecture: a five-category SDOH taxonomy with per-category definitions, few-shot exemplars, and a constrained output trigger; benchmarked against Llama-2-7B on CPU to document where smaller models break down on structured task specification.
- Ran a two-annotator manual evaluation over a stratified 109-record sample, reaching  $\kappa = 0.70$  (85.3% raw agreement); built an error taxonomy separating hallucination, prompt contamination, keyword anchoring, and missed implicit signals.
- Key finding: the model is well-calibrated at detecting absence of SDOH signal (91–98%), but its positive predictions are near-uniformly unsupported, defaulting to “Education Access and Quality” in 67% of consensus-incorrect labels — indicating lexical fallback rather than contextual inference.

### Research Assistant — Participatory AI Design with Home Care Cooperatives

University at Buffalo

Jan 2026 – Jul 2026

Buffalo, NY

Advised by Kenneth Joseph and Joy Ming — work led to CHI 2027 submission (see Publications)

- Co-designed and facilitated a multi-session workshop study with caregivers at member-owned home care cooperatives: pre-survey instrument, critical AI literacy curriculum, structured design activities, and post-test.
- Built the curriculum covering how AI systems are constructed and where they fail (algorithmic bias, hallucination, sycophancy bias, automation bias), mapping harms across individual, group, organizational, and systems levels alongside emerging responses (data cooperatives, Indigenous data sovereignty, community audit, collective bargaining).
- Applied BERTopic and TF-IDF analysis across open-ended survey responses on AI perception, anticipated benefits, and perceived risks to identify current use, desired changes, and workers’ foresight for the sector.
- Authored the resulting adoption framework for co-ops and NGOs: FATE-based tool evaluation, a stakeholder participation ladder, a data sensitivity tiering model, pilot-to-expand-or-exit criteria, and explicit refusal criteria for where AI should not be used (crisis support, conflict resolution, decisions affecting housing, benefits, or livelihood).

### Research Data Scientist — Perception of AI for Mental Health Support

University at Buffalo

Jul 2025 – Jul 2026

Buffalo, NY

Advised by Yotam Ophir

- Analyzed Reddit discourse on the use of AI for mental health support: how people describe turning to these tools, what they report gaining, and where they voice concern about dependence, misinformation, and substitution for human care.
- Code and analysis: [github.com/KhansaKhanam/AI4MentalHealth](https://github.com/KhansaKhanam/AI4MentalHealth)

## Research Assistant — Bayesian Optimization for Simulation Calibration

Institute of Artificial Intelligence and Data Science, University at Buffalo

Advised by Ashlee Ford Versypt

Sep 2024 – Jan 2026

Buffalo, NY

- Co-authored a constrained Bayesian optimization framework using Gaussian Process surrogates for both the objective (weighted RMSE against experimental invasion distance) and a morphological feasibility constraint, with a feasibility-informed acquisition function guiding the search.
- Replaced heuristic parameter tuning with a reproducible, uncertainty-aware calibration pipeline for agent-based models of collective cell invasion.

## Talks & Presentations

“Efficient Parameter Estimation in Agent-Based Models of Collective Cell Invasion via Gaussian Process Surrogates and Bayesian Optimization” — replacing heuristic parameter tuning in agent-based cancer invasion models with GP-surrogate Bayesian optimization under biological feasibility constraints.

- Talk, European Conference on Mathematical and Theoretical Biology (ECMTB 2026), University of Graz.
- Poster, SIAM Conference on the Life Sciences (LS26).
- Poster, Foundations of Computer-Aided Process Design (FOPAM 2026).
- Abstract accepted, Biomedical Engineering Society Annual Meeting (BMES 2026).

## Publications & Manuscripts

“Centering Care and Cooperation in the Design of AI Systems with Home Care Worker Cooperatives in the United States.” Joy Ming, Hawi Tolera, Omir Henriquez, Connor Wurst, Khansa Khanam Umar Sultan, Seventy Hall, Kenneth Joseph, and Geoffrey Gusoff. Under review, ACM CHI Conference on Human Factors in Computing Systems (CHI 2027).

*Manuscripts in preparation:*

- “Bayesian Optimization Approach for Agent-Based Modeling of Cancer Cell Migration and Invasion Phenotype.” Temitope O. Benson, Aneesh Krishna, Khansa Khanam Umar Sultan, and Ashlee N. Ford Versypt (Institute for Artificial Intelligence and Data Science, University at Buffalo). See Talks & Presentations for related conference activity.
- Multilingual evaluation of open-weight LLMs for Social Determinants of Health extraction from clinical notes (Rutgers SC&I).
- AI perception among Reddit users seeking mental health support — in preparation for journal submission (University at Buffalo).

## Industry Experience

### AI Data Engineer

Manage Your Business

May 2026 – Present

Buffalo, NY

- Architected end-to-end data infrastructure for an in-house ERP/CRM system (ticketing, attendance, payroll) on AWS, serving cross-functional teams across Saudi Arabia, India, and Pakistan; designed the relational schema (30–45 entities, 5 modules) and a separate star-schema for BI consumption.
- Built and maintained Airflow DAGs orchestrating the ticketing pipeline (scheduling, retries, backfills across 30+ daily runs) and the dbt transformation layer feeding downstream analytics.
- Ran ETL from unstructured sources (spreadsheets, attorney-drafted lease PDFs) into a normalized entity model supporting a commercial real-estate underwriting system; used Databricks for workforce analytics across 400+ employee records and built the 5-year pro forma and DSCR analysis for an investor-facing capital raise.

### Business Intelligence Analyst

SafetyConnect

Aug 2021 – Dec 2023

India

- Benchmarked 12+ competitors and built demand-forecasting dashboards across 3 verticals, informing a US market-entry strategy supporting a \$1.2M–\$2M Seed A raise.
- Synthesized 20+ stakeholder interviews into a feature-gap analysis presented at 8+ C-suite reviews, accelerating launch readiness by 30%.

## Technical Skills

- **Languages & Tools:** Python, SQL, AWS, Databricks, dbt, Airflow, Tableau, Git
- **ML / NLP:** Fine-tuning and evaluation of open-weight LLMs (Llama-2/3, Bode, Flan-T5, text-bison), in-context learning, BERTopic, SGDClassifier, SHAP, prompt architecture design
- **Research Methods:** Mixed methods (qualitative + quantitative), workshop/participatory design studies, survey design, inter-annotator agreement (Cohen’s  $\kappa$ ), Bayesian optimization, Gaussian Process surrogate modeling

## Honors & Awards

- Grace Hopper Celebration Scholarship Recipient, AnitaB.org, 2024
- Finalist, AI21 Labs Hackathon